One-electron X2C: decoupling, the effective Hamiltonian, and the renormalization

Equations behind `X2C_DHF/X2C_hamiltonian.{h,cpp}` and the keyword `X2C_NORM`

1. Basis changes of a generalized eigenproblem

Everything below is one identity applied twice. Let $H\,C = S\,C\,e$ be a generalized eigenproblem and U any matrix (it need *not* be unitary, nor even square). Writing $C = U\,C_1$ and multiplying from the left by $U^\dagger$ gives

$$H_1 C_1 = S_1 C_1\, e, \qquad H_1 = U^\dagger H\, U, \quad S_1 = U^\dagger S\, U. \tag{1}$$

The eigenvalues e are unchanged. For a square invertible U the problems are equivalent; for a rectangular U (more rows than columns) Eq. (1) is a problem in the subspace spanned by the columns of U. What changes is the *metric* S_1: it is the identity only if $U^\dagger S\, U = 1$. This is the whole point of the renormalization of Sec. 4.

2. Elimination of the small component

In the restricted-kinetic-balance basis (see `doc/RKB.tex`) the one-electron Dirac problem in the orthonormalizable [Large; Small] AO basis is $H\,C = S\,C\,e$ with

$$H = \begin{pmatrix} H_{LL} & H_{LS} \\ H_{SL} & H_{SS} \end{pmatrix}, \qquad S = \begin{pmatrix} S_{LL} & 0 \\ 0 & S_{SS} \end{pmatrix}. \tag{2}$$

Every positive-energy eigenvector $C = (C_L; C_S)$ satisfies a *linear relation* between its components,

$$C_S = X\, C_L, \qquad X = C_S\, C_L^{-1} \tag{3}$$

(the same X for every positive-energy eigenvector; in the code `X2CHamiltonianResult::r_matrix` is this X, *not* the R of Sec. 4). Take the rectangular $U = \begin{pmatrix} 1 \\ X \end{pmatrix}$ in Eq. (1):

$$H_1 = U^\dagger H\, U = H_{LL} + H_{LS}X + (H_{LS}X)^\dagger + X^\dagger H_{SS} X \;\equiv\; h_{x2c}, \tag{4}$$

$$S_1 = U^\dagger S\, U = S_{LL} + X^\dagger S_{SS} X \;\equiv\; \Lambda, \tag{5}$$

$$h_{x2c}\, C_L = \Lambda\, C_L\, e. \tag{6}$$

This $n \times n$ problem contains *exactly* the positive-energy eigenvalues of the $2n \times 2n$ Dirac problem (no approximation: this is the exact, one-electron X2C, the "X2C-1e" of Ilias and Saue). It is verified in `main.cpp`'s X2C report: $\max|E_{X2C} - E_{DHF}| \approx 10^{-10}$ Ha.

Meaning of Λ. The 4-component eigenvectors are normalized in the full metric, $C_L^\dagger S_{LL} C_L + C_S^\dagger S_{SS} C_S = 1$, i.e. $C_L^\dagger \Lambda C_L = 1$. $\Lambda = S_{LL} + X^\dagger S_{SS} X$ exceeds the plain overlap S_{LL} of the large AOs by the *weight of the small component* ($S_{SS} \sim T/2c^2$, so $\Lambda - S_{LL}$ is of order $(v/c)^2$ and largest for core orbitals of heavy atoms).

3. Why Eq. (6) is not yet usable in a Hartree–Fock code

An SCF code evaluates densities and two-electron contractions assuming orbitals that are orthonormal in the *AO overlap*, $C^\dagger S_{LL} C = 1$, and uses nonrelativistic two-electron integrals over the same AO functions (no two-electron picture change). The orbitals of Eq. (6) instead satisfy $C_L^\dagger \Lambda C_L = 1$. Using h_{x2c} together with the metric S_{LL} (orbitalizing with $S_{LL}^{-1/2}$, as an earlier version of this code did) is inconsistent: the Hamiltonian belongs to the metric Λ. For Ne/cc-pVDZ this error is 0.137 Ha in the Hartree–Fock energy (Sec. 6).

4. Renormalization: a second basis change

Apply Eq. (1) once more, now with a *square* invertible R ($n \times n$) acting on $C_L = R\, C_2$:

$$H_2 = R^\dagger h_{x2c}\, R, \qquad S_2 = R^\dagger \Lambda\, R, \qquad H_2 C_2 = S_2 C_2\, e. \tag{7}$$

Choose R such that the new metric is the plain overlap,

$$R^\dagger \Lambda\, R = S_{LL} \qquad \Longrightarrow \qquad h_{FW}\, C_2 = S_{LL}\, C_2\, e, \quad C_2^\dagger S_{LL} C_2 = 1, \quad h_{FW} \equiv R^\dagger h_{x2c} R. \tag{8}$$

The eigenvalues e are still those of the Dirac problem, and the orbitals $C_2 = R^{-1} C_L$ are orthonormal in S_{LL}. h_{FW} is the Hamiltonian the SCF uses (`X2C_NORM TRUE`).

The symmetric choice. $R^\dagger \Lambda R = S_{LL}$ does not fix R: $R \to RV$ with $V^\dagger S_{LL} V = S_{LL}$ gives the same eigenvalues, but mixes the AOs, which would spoil the two-electron integrals defined for the original functions. The standard choice is the one closest to the identity,

$$\boxed{R = S_{LL}^{-1/2} \left(S_{LL}^{-1/2}\, \Lambda\, S_{LL}^{-1/2} \right)^{-1/2} S_{LL}^{1/2}} \tag{9}$$

with Hermitian matrix powers (diagonalization; eigenvalues of S_{LL} below 10^{-14} are dropped). Check: with $M = S_{LL}^{-1/2} \Lambda S_{LL}^{-1/2}$, $R^\dagger \Lambda R = S_{LL}^{1/2} M^{-1/2} \left(S_{LL}^{-1/2} \Lambda S_{LL}^{-1/2} \right) M^{-1/2} S_{LL}^{1/2} = S_{LL}^{1/2} M^{-1/2} M M^{-1/2} S_{LL}^{1/2} = S_{LL}$. To first order $R \simeq 1 - \frac{1}{2} S_{LL}^{-1} (\Lambda - S_{LL})$, i.e. R removes the small-component weight.

The two steps as one transformation. $U_{\mathrm{tot}} = \begin{pmatrix} 1 \\ X \end{pmatrix} R$ ($2n \times n$): $h_{FW} = U_{\mathrm{tot}}^\dagger H\, U_{\mathrm{tot}}$ and $U_{\mathrm{tot}}^\dagger S\, U_{\mathrm{tot}} = S_{LL}$.

5. Implementation

- `buildX2CHamiltonian` (`X2C_hamiltonian.cpp`): X, Λ, h_{x2c} of Eqs. (4)–(5), and $h_{x2c}^{\mathrm{ortho}} = \Lambda^{-1/2\dagger} h_{x2c} \Lambda^{-1/2}$, whose eigenvalues are the exact positive-energy spectrum (the report's check).

- `renormalizedX2CHamiltonian`: $h_{FW} = R^\dagger h_{x2c} R$ with R of Eq. (9). In `main.cpp` it replaces `x2c_hamiltonian.h_x2c` right after construction when `X2C_NORM TRUE` (default); every later use (SCF, functional stage, restart, UCCSD) then sees h_{FW}.
- The SCF orthogonalizes with the large-component block of the RKB orthonormalization matrix, i.e. with $S_{LL}^{-1/2}$ (or LOWGEN), which is the correct metric for h_{FW}. With `X2C_NORM FALSE` the raw h_{x2c} is used in the same orthogonalization (old behaviour: not a consistent X2C; its eigenvalues are only close to the exact ones, the report prints the deviation).
- Two-electron integrals are the ordinary nonrelativistic ones over the large-component AOs; no picture-change correction is applied to them or to the density. Densities and natural orbitals (WFN files, RDMs) are the nonrelativistic form of the density built from C_2 and S_{LL}. A density with picture change would use $C_L = R\,C_2$ plus a small-component part, and is not implemented.

6. Validation

Ne, cc-pVDZ (spherical), point nucleus, $c = 137.036$, X2C-HF energy (Hartree):

`X2C_NORM TRUE` (this code)	−128.6328725
PySCF spinor X2C-1e (contracted decoupling)	−128.6328725
`X2C_NORM FALSE` (raw h_{x2c})	−128.7698416
4-component DHF (this code and PySCF)	−128.6318

Water, STO-3G: total −75.0057781 (this code) vs. −75.0057780 (PySCF). Forcing $R = 1$ in PySCF's own `sfx2c1e` reproduces the old value (−128.7698325 against −128.7698416 here), which identifies the missing renormalization as the entire discrepancy. X2C-HF lies 1.0 mHa above DHF for Ne, the usual size of the missing two-electron picture-change error.